PES UNIVERSITY

Department of Computer Science & Engineering

Digital Forensics and Incident Responses

UE23CS342BA6

Assignment 5 Submission

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Section	G
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Campus	RR

Topic: Lab 5: Network Forensics

UE23CS342BA6

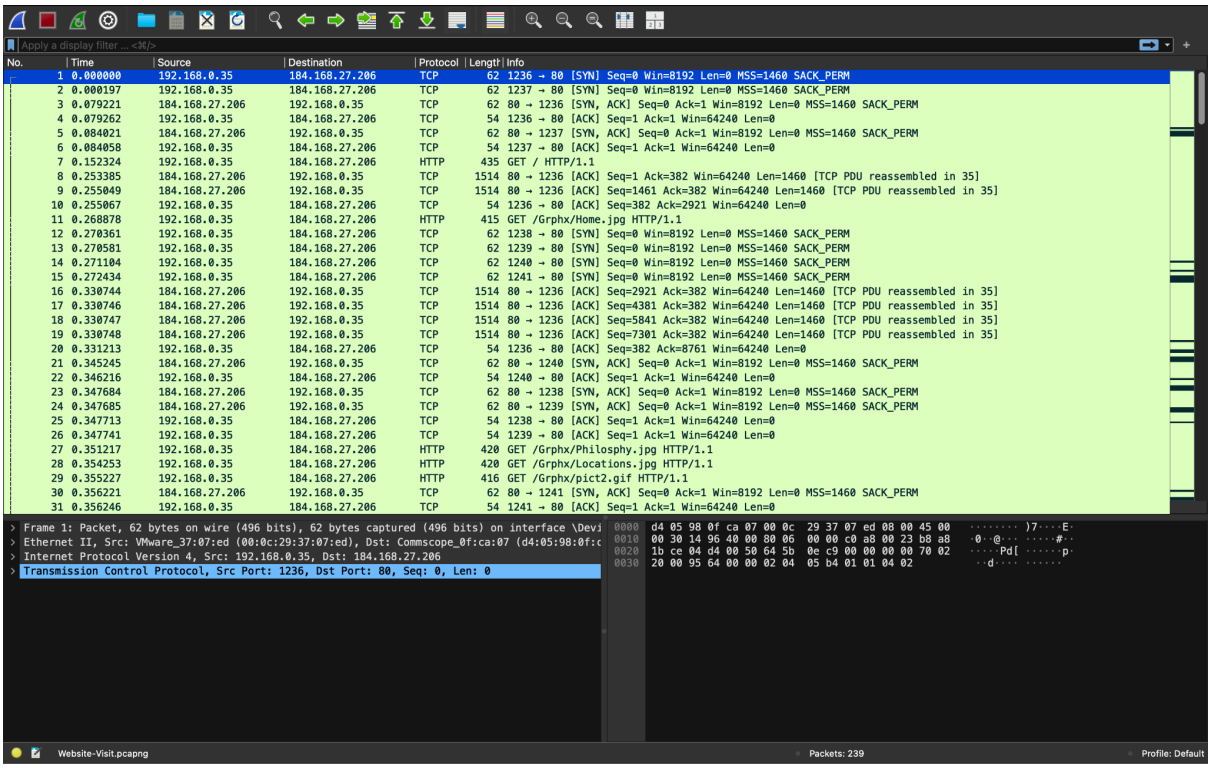
Objective:

The objective of this experiment is to understand the fundamentals of network forensics by analyzing captured network traffic using forensic tools. Students will learn how to examine packet capture files, identify communication between hosts, and reconstruct artifacts transmitted over the network.

Part A: Network Traffic Analysis: Host Foot-printing and File Extractions

Step1: Download the following packet capture: “Website-Visit.pcapng.”

1) Launch wireshark and open the file

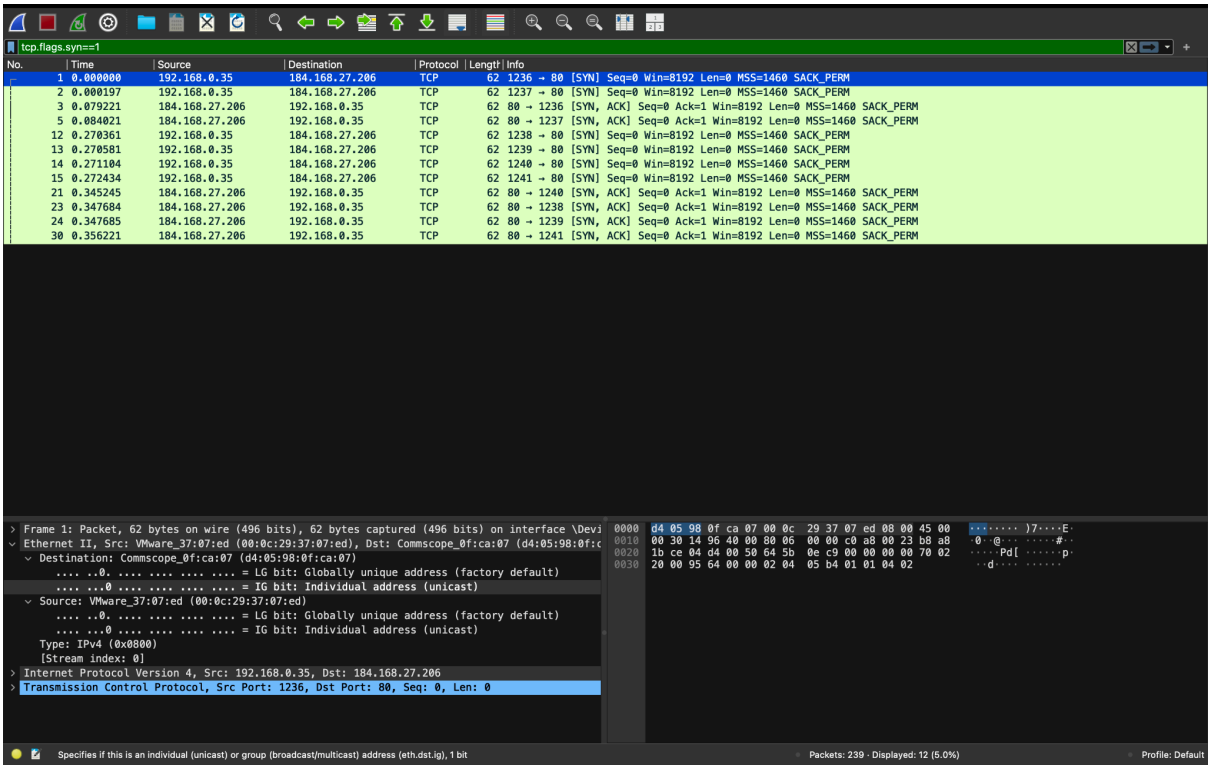


Explanation:

The packet capture file contains recorded network traffic from earlier communications between a client and a web server. It is used to study network activity and extract digital forensic evidence.

Question 1. What is the IP address of the visited web site?

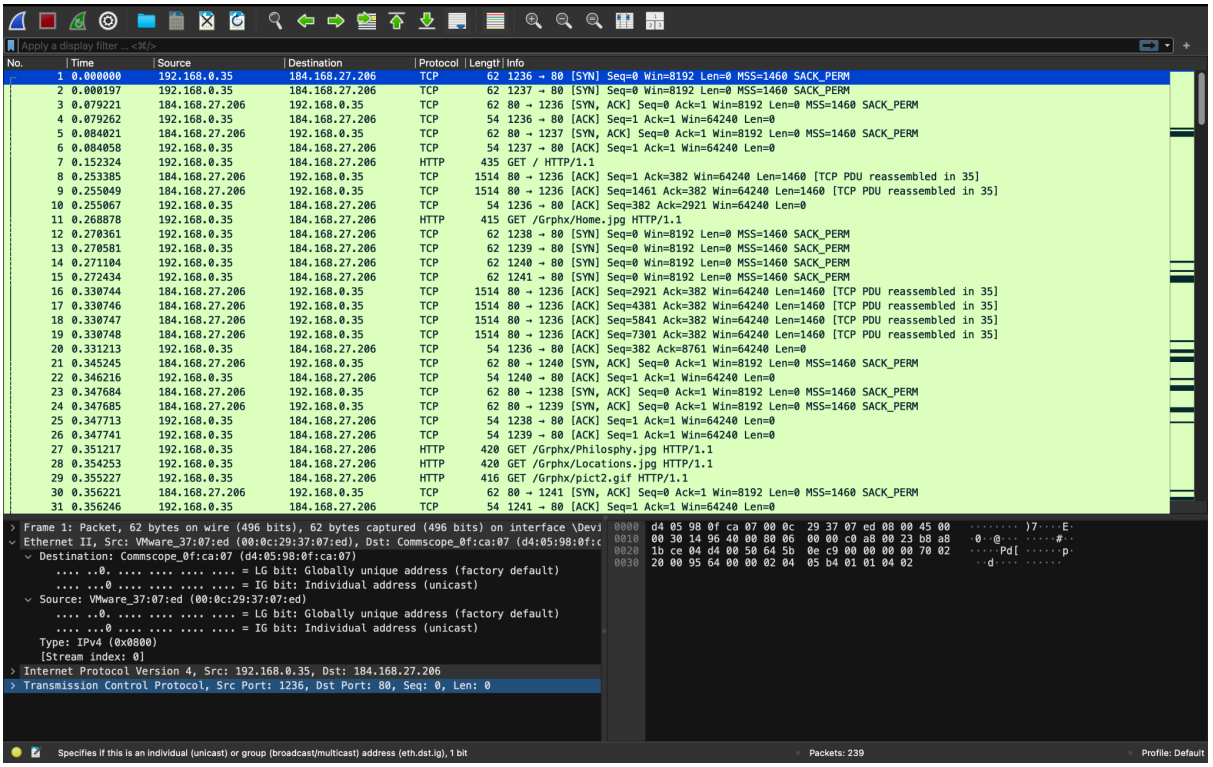
In the textbox above the packet information tabs enter the following text to filter the results looking for packets with the SYN flag set: tcp.flags.syn==1



Answer: As we can see above IP of visited site is: 184.168.27.206

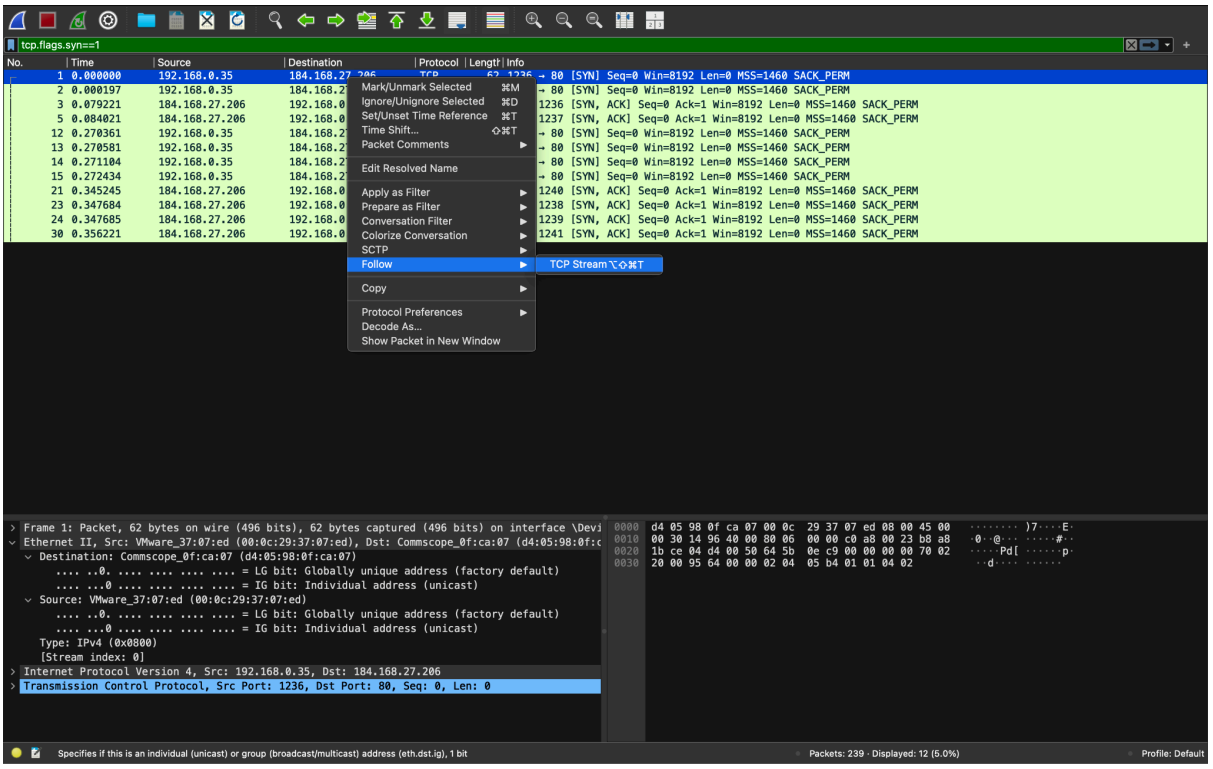
Explanation:
TCP SYN packets are used to start a TCP connection. By filtering packets that have the SYN flag set, we can detect the first communication between the client and the web server and identify the destination IP address of the website that was accessed.

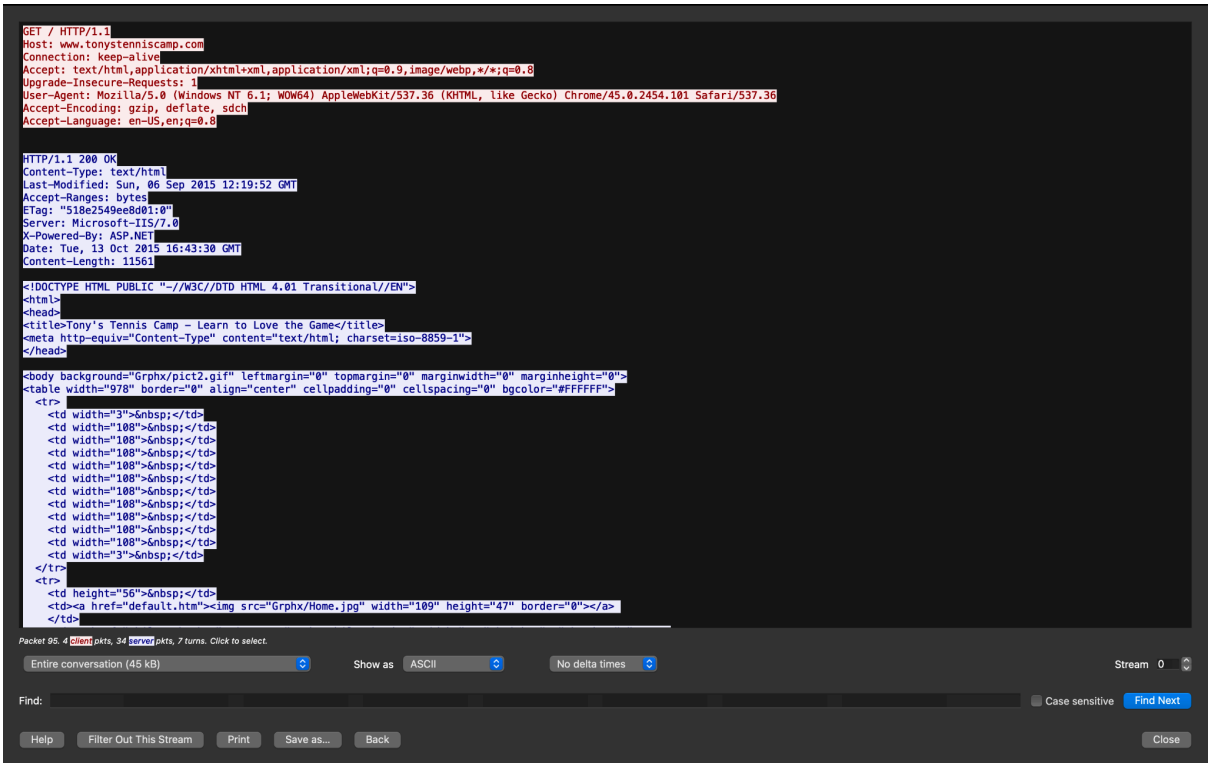
Question 2: On what port is the web server listening?



Answer: Web server port: 80 as we can see in the TCP SYN packet

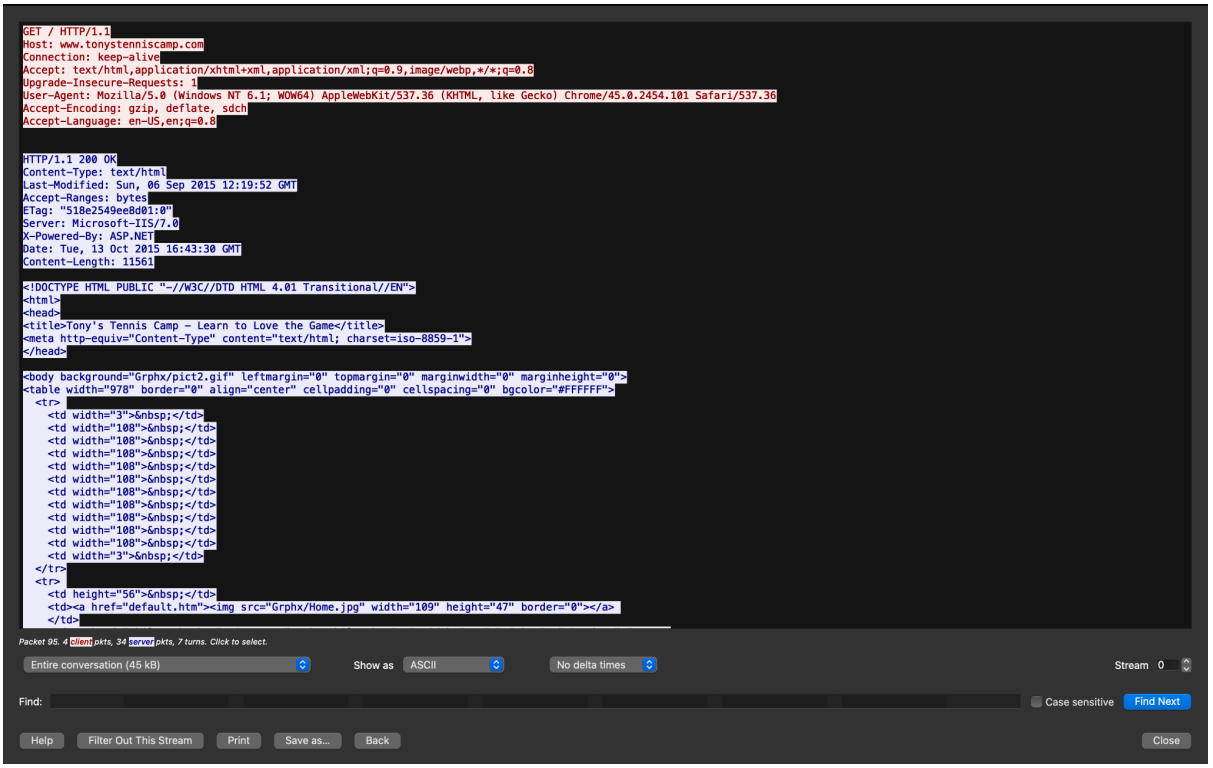
Question3: What is the name of the web site, which was visited?
In the Packet List Pane, right-click on the packet numbered 1 and from the pull-down menu select “Follow TCP Stream” as shown below.





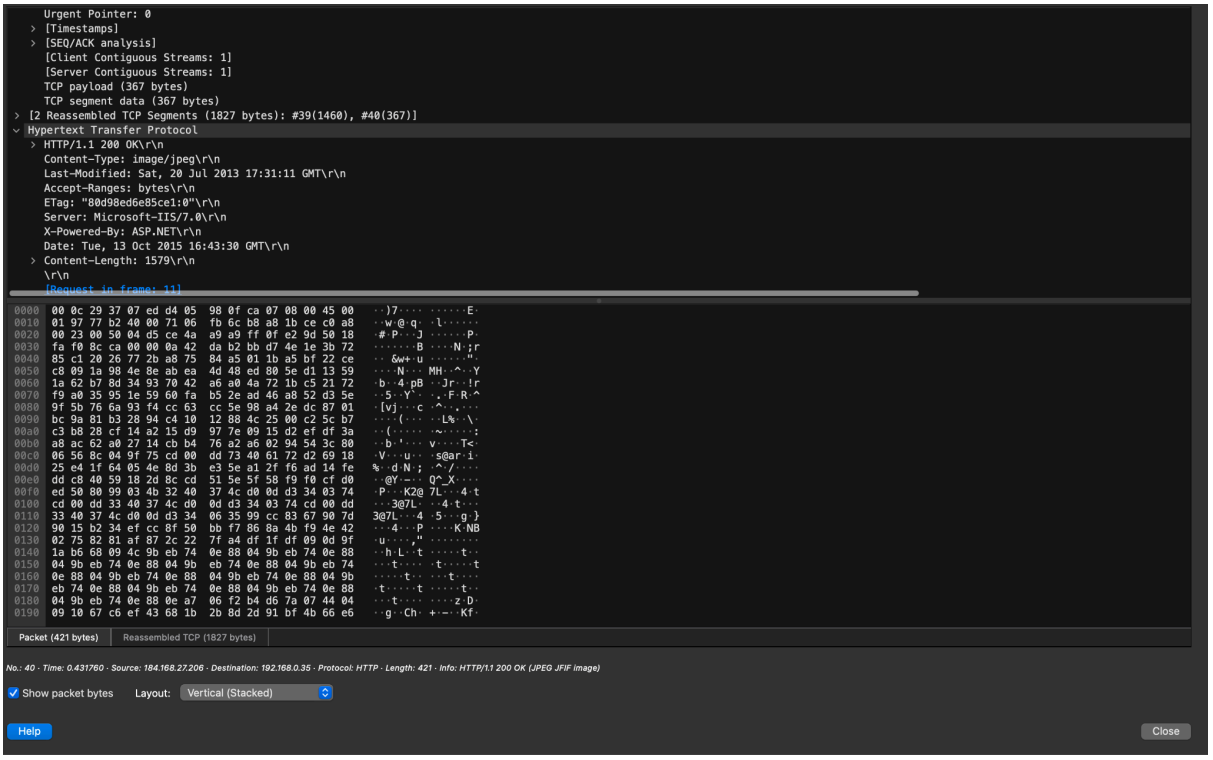
Answer: The website name is: www.tonystenniscamp.com .The HTTP Host header specifies the domain name of the web server that the client is requesting access to.

Question4: Was the connection to the web site successful from the server's point of view?



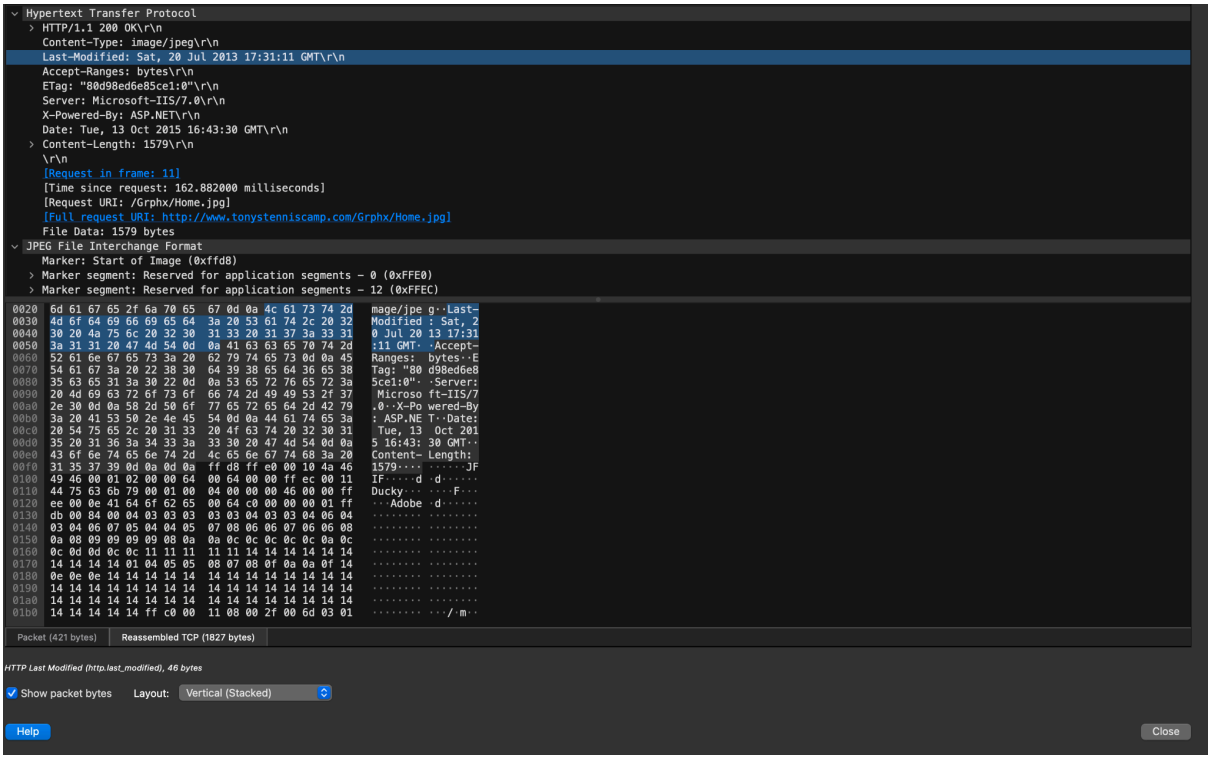
Answer: The connection was successful. The HTTP status code 200 OK indicates that the server successfully handled the request and delivered the requested content.

Question5: What kind of web platform is being used to host the server?



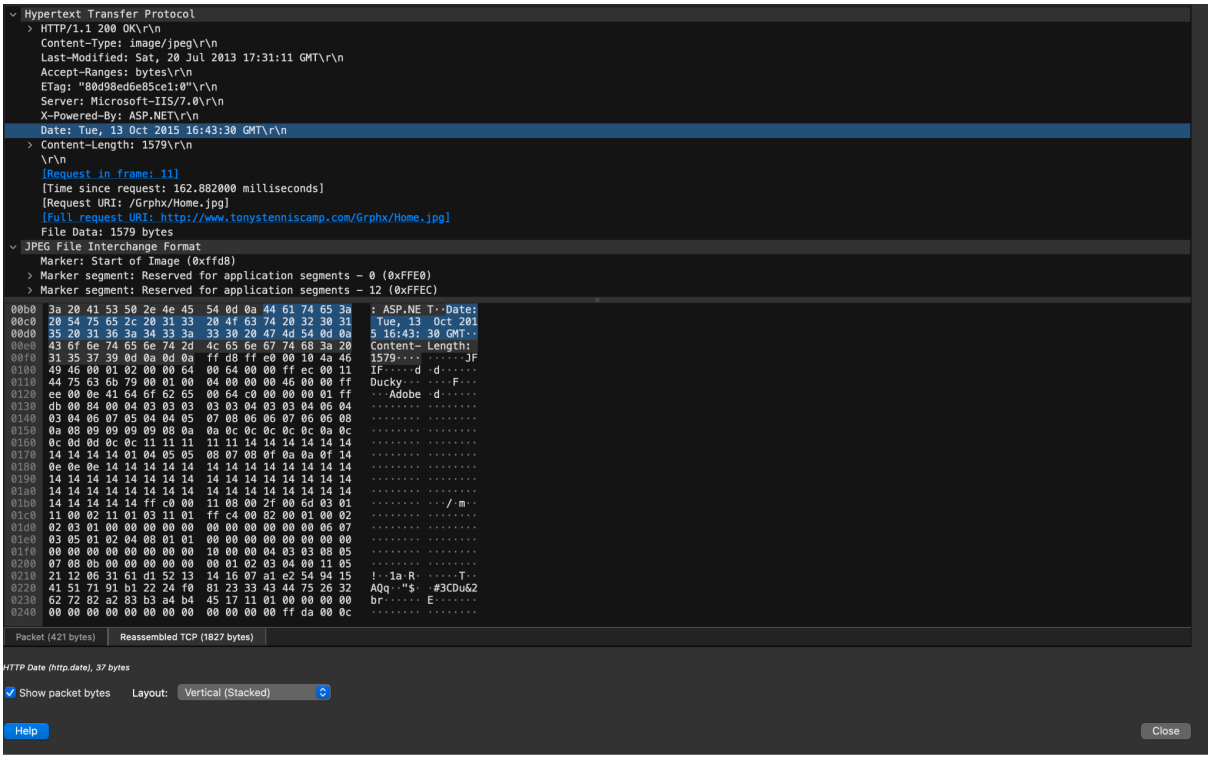
Answer: The web platform is Microsoft-IIS/7.0 The Server header in an HTTP response indicates the web server software and version that is hosting the website.

Question6: When was this visited web page last updated by the web author/provider?



Answer: As we can see above, the last updated date: Sun, 06 Sep 2015 12:19:52 GMT. The Last-Modified HTTP header specifies the date and time when the requested resource was most recently updated on the web server.

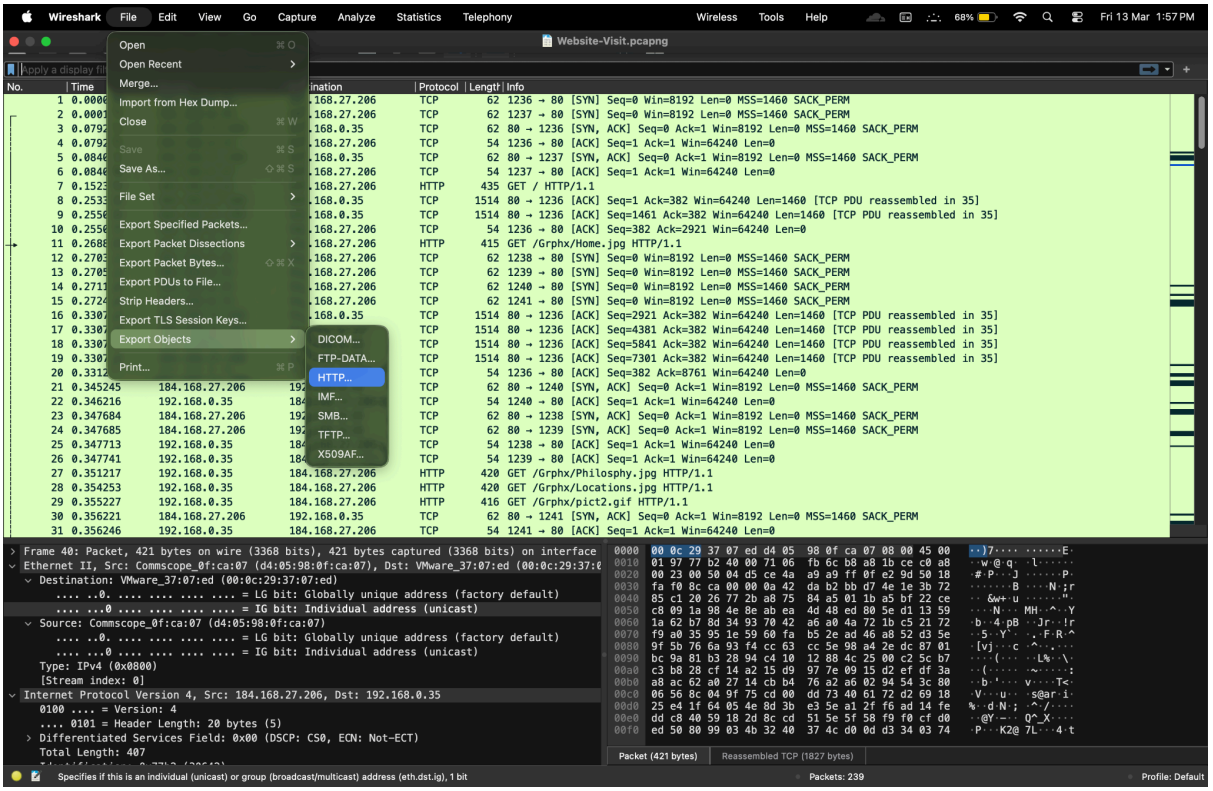
Question7: When was this particular web page accessed?

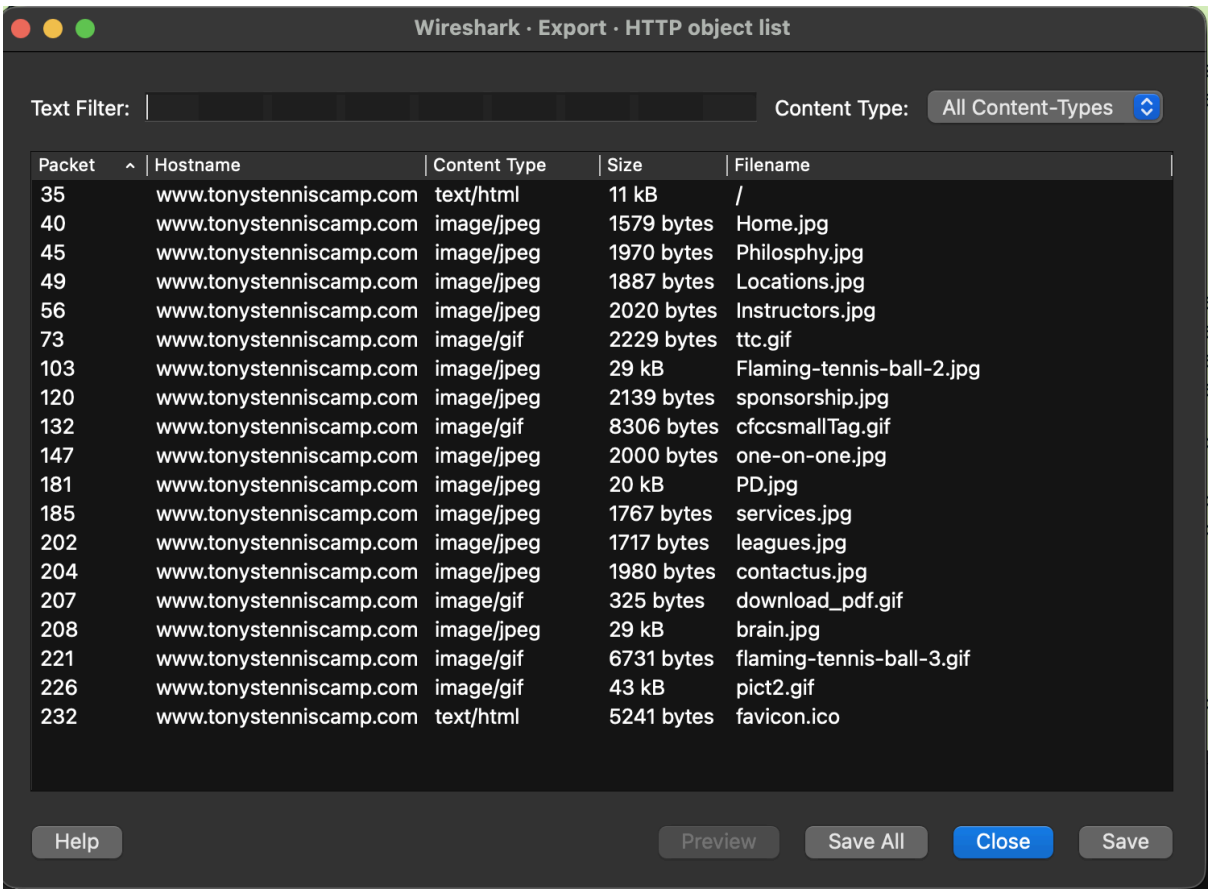


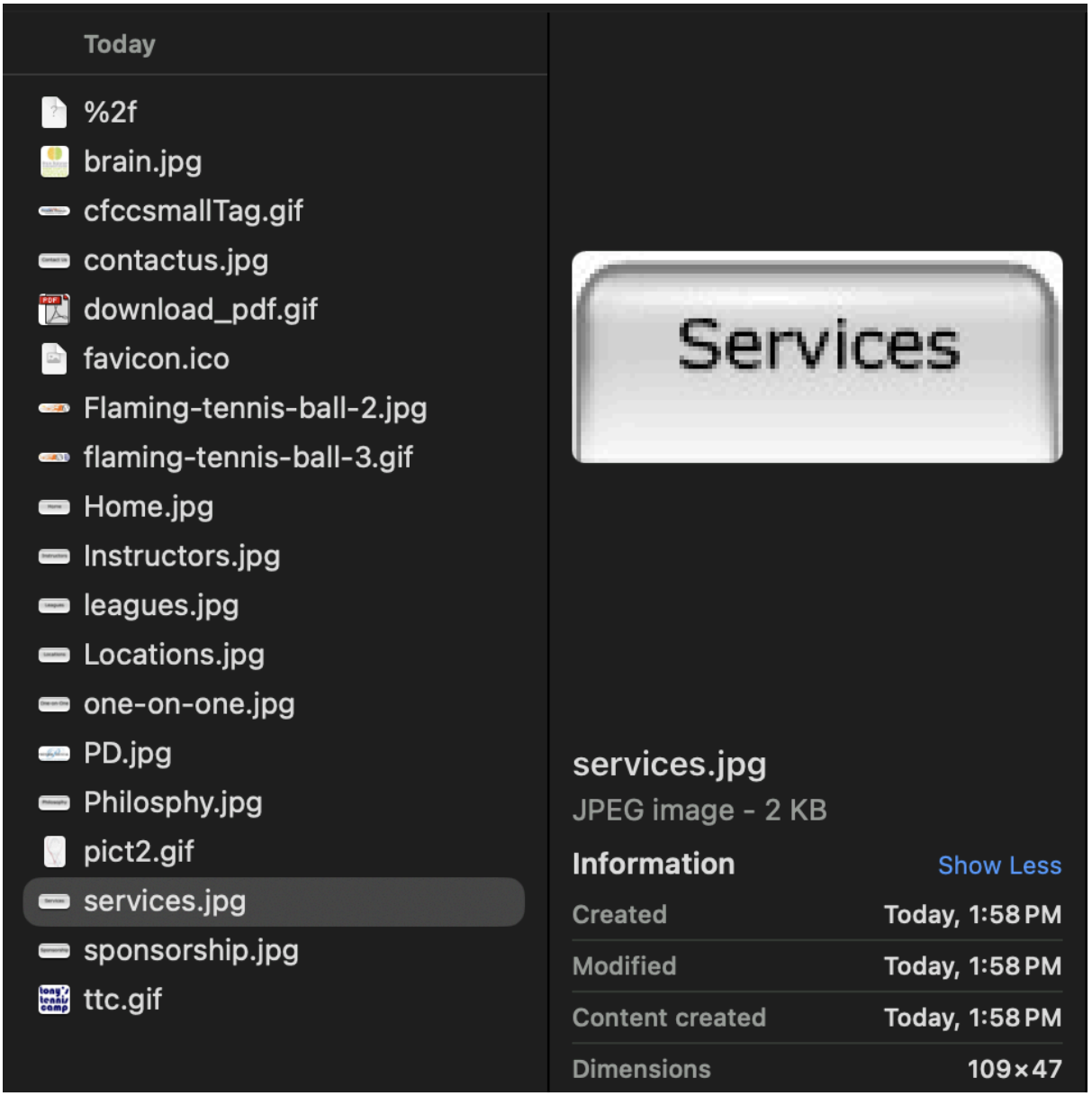
Answer: As we can see above, the page accessed time is Tue, 13 Oct 2015 16:43:30 GMT. The Date HTTP header shows the time at which the server generated the HTTP response, indicating when the page was accessed.

Step2: Recover artifacts from the network packet capture,

2) Go to the main menu, select “File” , select “Export objects,” and then select “HTTP”).



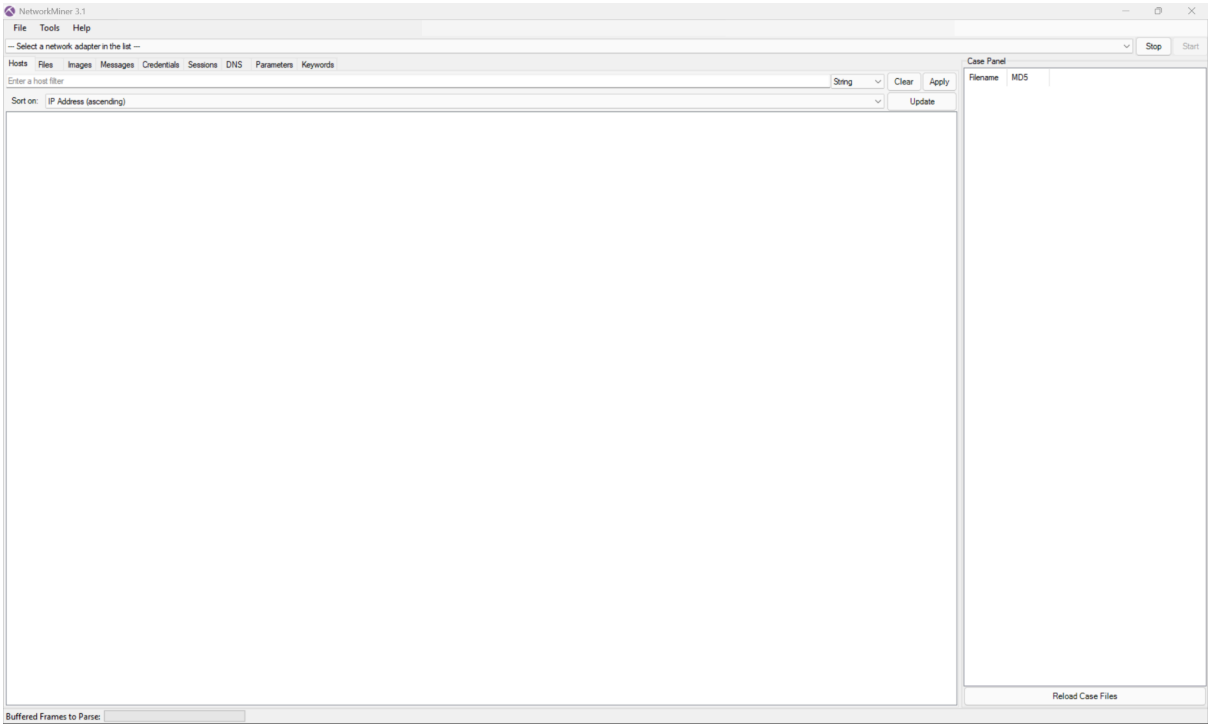

Wireshark · Export · HTTP object list
Text Filter:
Content Type:
All Content-Types
Packet
Hostname
Content Type
Size
Filename
35 www.tonystenniscamp.com text/html 11 kB /
40 www.tonystenniscamp.com image/jpeg 1579 bytes Home.jpg
45 www.tonystenniscamp.com image/jpeg 1970 bytes Philosphy.jpg
49 www.tonystenniscamp.com image/jpeg 1887 bytes Locations.jpg
56 www.tonystenniscamp.com image/jpeg 2020 bytes Instructors.jpg
73 www.tonystenniscamp.com image/gif 2229 bytes ttc.gif
103 www.tonystenniscamp.com image/jpeg 29 kB Flaming-tennis-ball-2.jpg
120 www.tonystenniscamp.com image/jpeg 2139 bytes sponsorship.jpg
132 www.tonystenniscamp.com image/gif 8306 bytes cfccsmallTag.gif
147 www.tonystenniscamp.com image/jpeg 2000 bytes one-on-one.jpg
181 www.tonystenniscamp.com image/jpeg 20 kB PD.jpg
185 www.tonystenniscamp.com image/jpeg 1767 bytes services.jpg
202 www.tonystenniscamp.com image/jpeg 1717 bytes leagues.jpg
204 www.tonystenniscamp.com image/jpeg 1980 bytes contactus.jpg
207 www.tonystenniscamp.com image/gif 325 bytes download_pdf.gif
208 www.tonystenniscamp.com image/jpeg 29 kB brain.jpg
221 www.tonystenniscamp.com image/gif 6731 bytes flaming-tennis-ball-3.gif
226 www.tonystenniscamp.com image/gif 43 kB pict2.gif
232 www.tonystenniscamp.com text/html 5241 bytes favicon.ico
Help
Preview
Save All
Close
Save

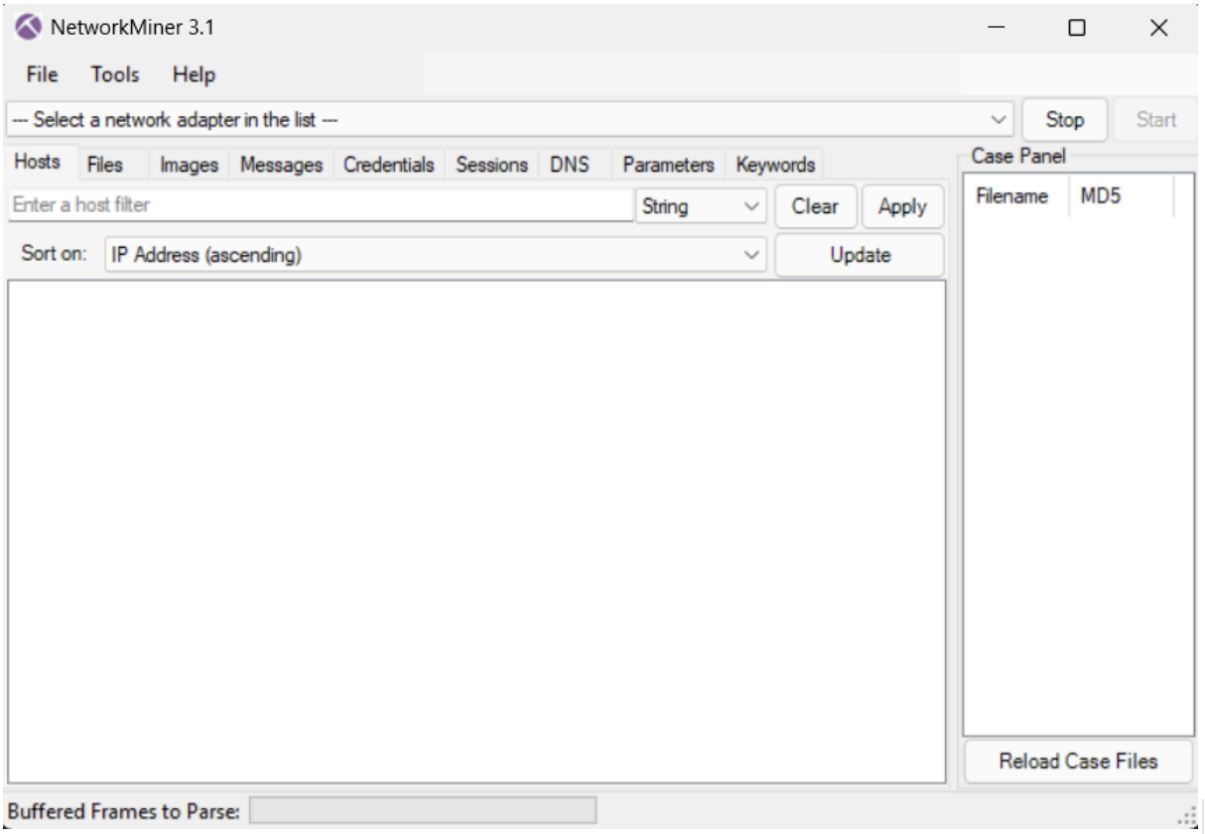


Part B: Network Artifact Reconstruction using NetworkMiner

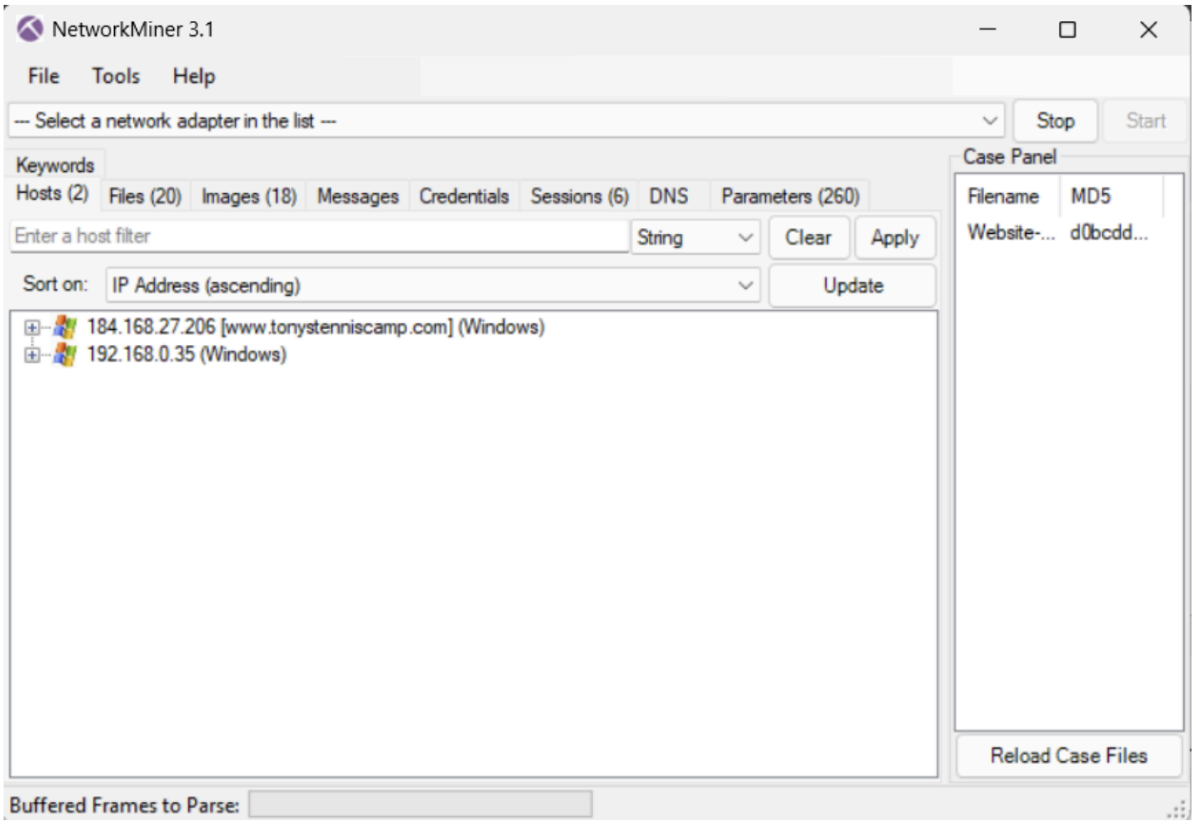
In this activity you will use NetworkMiner to analyze the same packet capture and automatically reconstruct artifacts such as hosts, files, images, and sessions.

Step1: Download and extract NetworkMiner from the official website and launch it





Step2: From the menu select File → Open. This opens the previously used packet capture file. NetworkMiner will automatically begin analyzing the network traffic.



Step3: Host analysis

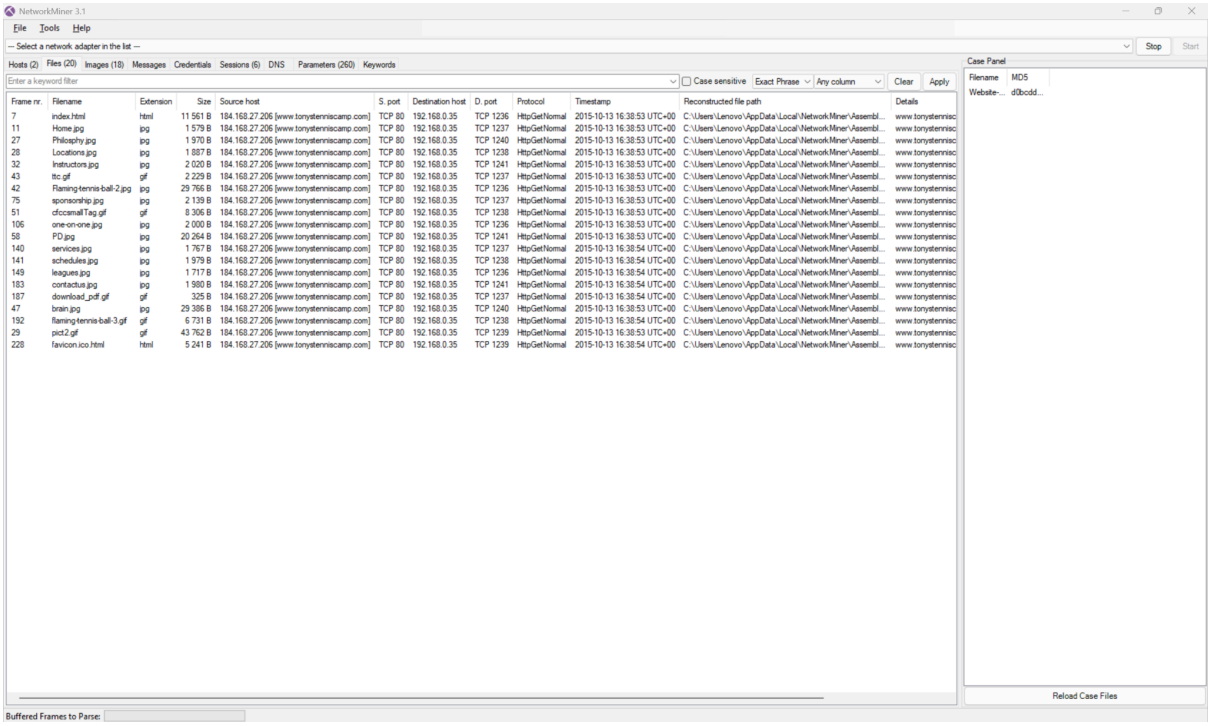
3) Open the Hosts tab. This tab displays the systems detected in the network communication.

Question 1. What is the IP address of the web server identified in the Hosts tab?
Answer: The IP address of the web server is 184.168.27.206.

Explanation: The Hosts tab in NetworkMiner displays all systems participating in the captured traffic. In this case, two hosts were observed: a local client (192.168.0.35) and a remote server (184.168.27.206). The remote server is linked to the domain www.tonystenniscamp.com and has a public IP address, indicating that it is the web server hosting the requested webpage.

Step4: Files reconstruction

4) Open the Files tab. NetworkMiner automatically reconstructs files transferred through the network.



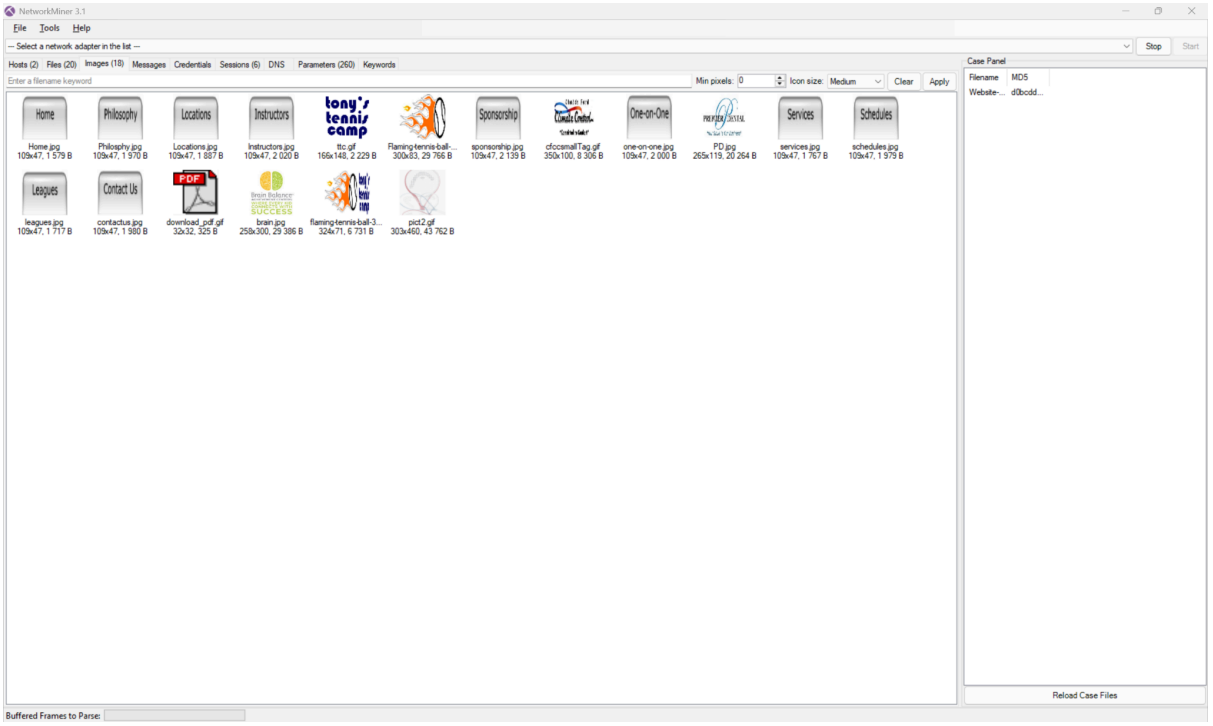
Question 2. How many files were reconstructed from the packet capture?
Answer: As we can see above, 20 files were captured

Question 3. What types of files were extracted?
Answer: As we can observe the captured file types to be - html, jpg and gif

Explanation:
NetworkMiner rebuilds transferred files by piecing together network packets associated with HTTP sessions.

Step5: Image reconstruction

5) Open the Files tab. NetworkMiner reconstructs images from HTTP traffic.

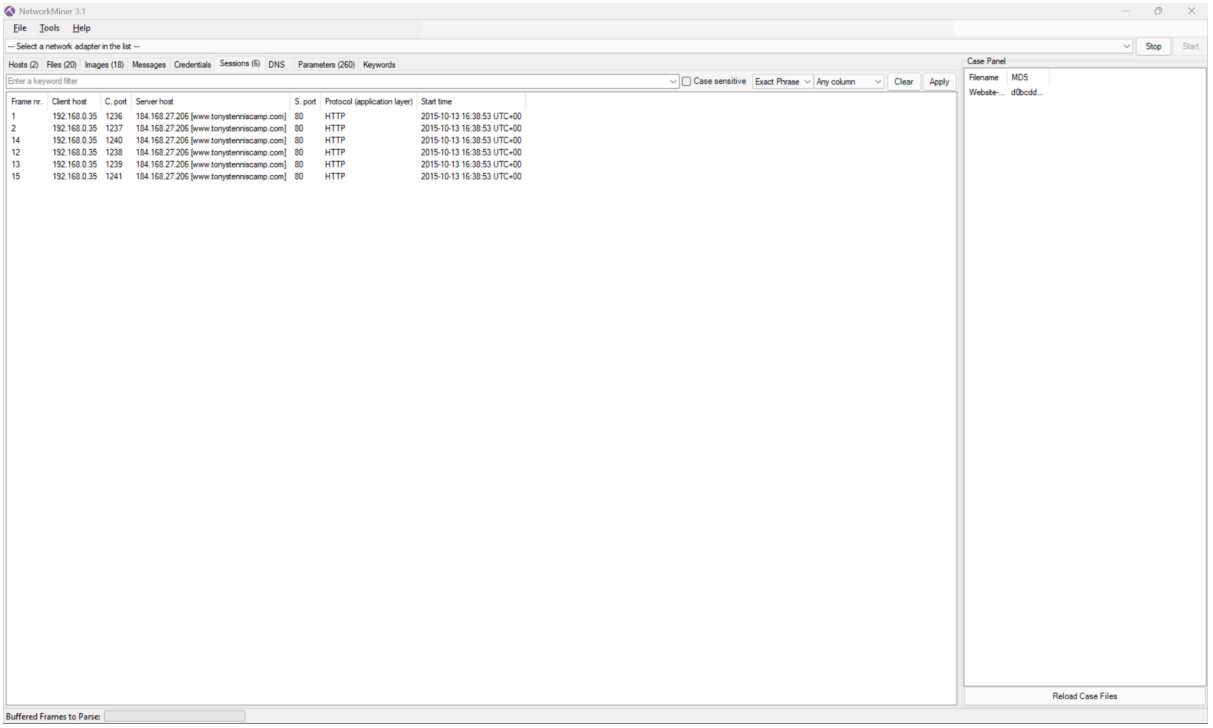


Question 4. How many images were reconstructed from the network traffic?
Answer: As per the above screenshot, we are able to reconstruct 18 images

Explanation:
Images transferred through HTTP traffic are reconstructed by analyzing HTTP responses containing image content.

Step6: Session Analysis

6) Open the Sessions tab. This tab displays communication sessions between hosts.

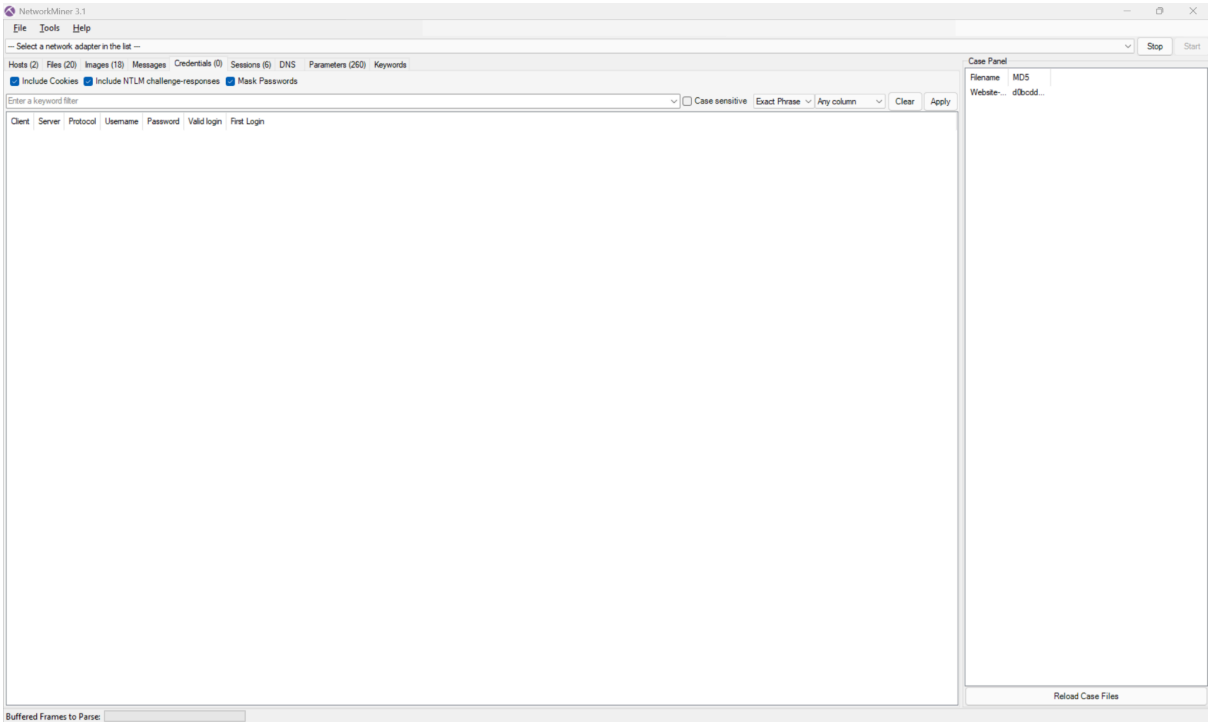


Question 5. Which protocol is used for the communication session between the client and server?
Answer: We can see that HTTP protocol is used for the communication session between the client and server.

Explanation:
The Sessions tab shows the communication exchanges between hosts along with the protocols used during the network activity.

Step7: Credential Design

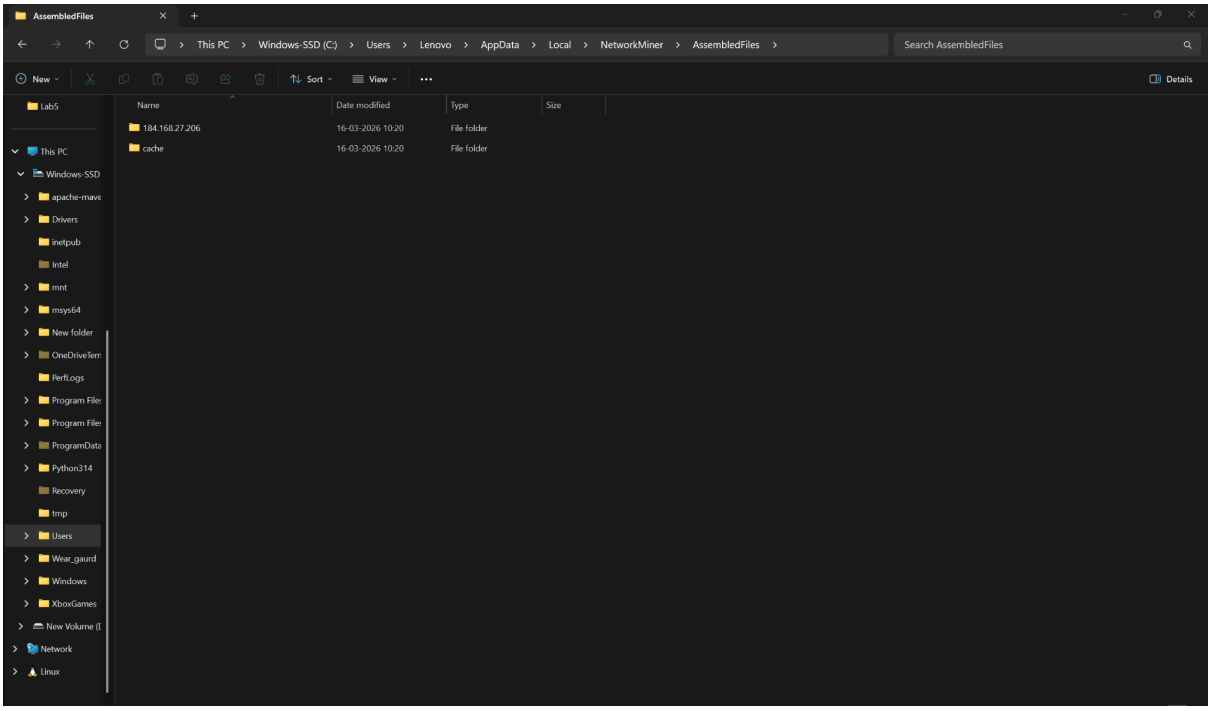
7) Open the Credentials tab. NetworkMiner may extract authentication data if present.



Question 6. Were any credentials detected in the packet capture?
Answer: No, there were no credentials detected in the packet capture

Step8: Export reconstructed files

8) Open folder: C:\UsersLenovo\AppData\Local\NetworkMiner\AssembledFiles



```
C:\Users\Lenovo\AppData\Local\NetworkMiner\AssembledFiles>tree /F
Folder PATH listing for volume Windows-SSD
Volume serial number is A626-409F
C:.
├───184.168.27.206
│   └───TCP-80
│       │   favicon.ico.html
│       │   index.html
│       │
│       └───Grphx
│               brain.jpg
│               cfccsmallTag.gif
│               contactus.jpg
│               download_pdf.gif
│               Flaming-tennis-ball-2.jpg
│               flaming-tennis-ball-3.gif
│               Home.jpg
│               Instructors.jpg
│               leagues.jpg
│               Locations.jpg
│               one-on-one.jpg
│               PD.jpg
│               Philosphy.jpg
│               pict2.gif
│               schedules.jpg
│               services.jpg
│               sponsorship.jpg
│               ttc.gif
│
└───cache

C:\Users\Lenovo\AppData\Local\NetworkMiner\AssembledFiles>
```

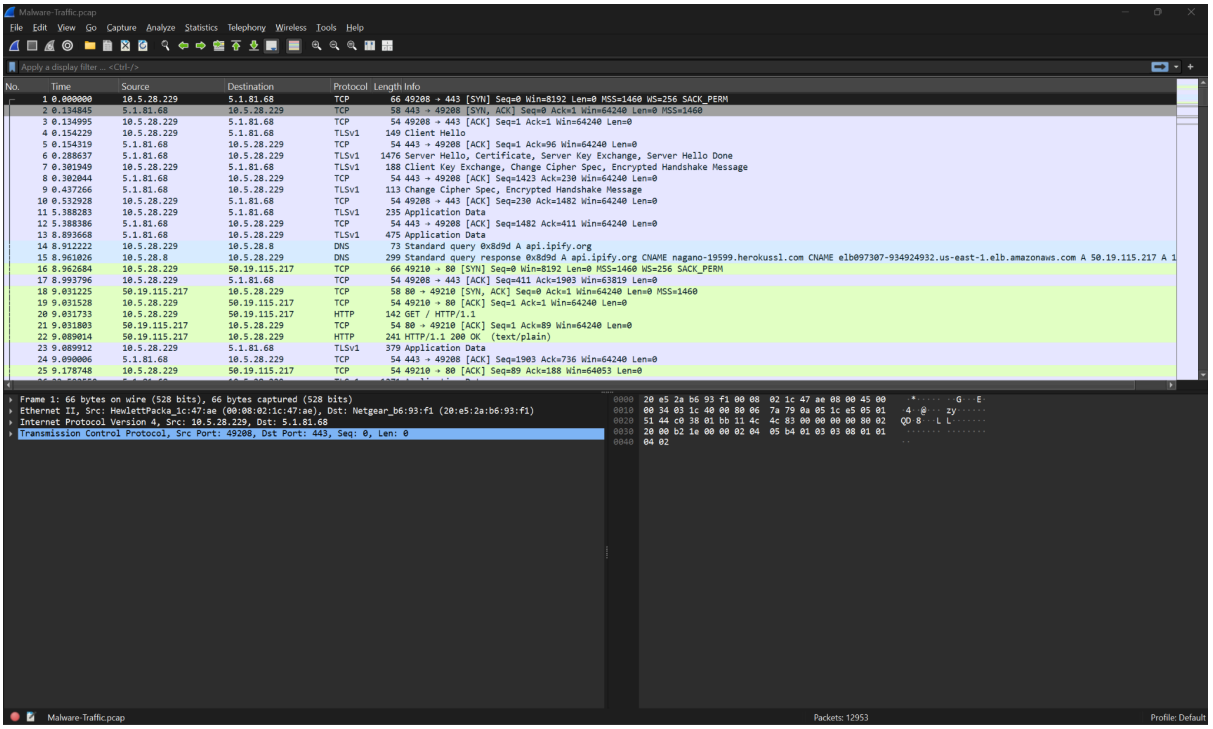
Part C: Malware traffic analysis

For this lab we will be taking a look at the network traffic sent by systems infected with a malware known as Trickbot. Trickbot is often distributed through malspam. Emails from these campaigns contain links to download malicious files disguised as invoices or documents. These files may be Windows executable files for Trickbot, or they may be some sort of downloader for the Trickbot executable. In some cases, links from these emails return a zip archive that contains a Trickbot executable or downloader.

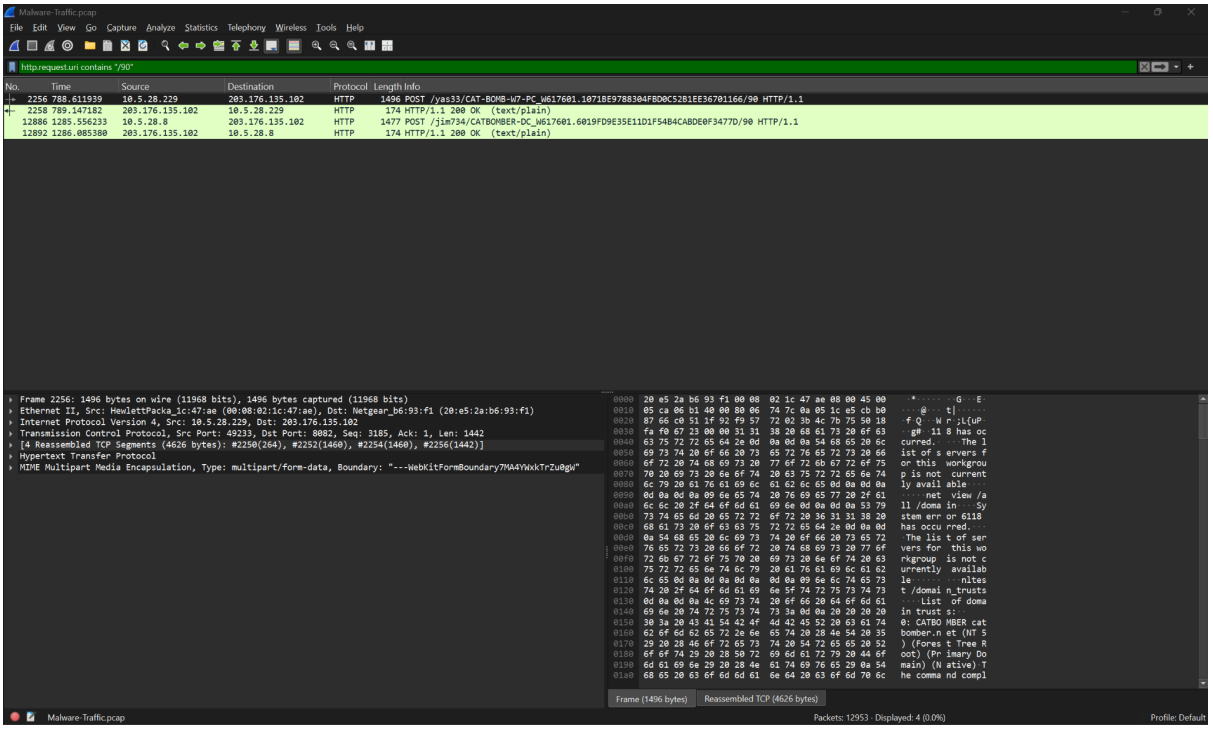
This Trickbot infection is in an Active Directory (AD) environment where the infection spreads to the Domain Controller (DC).

Step1: Download the following packet capture: MalwareTraffic-1.zip. This zip is encrypted and the password for it is “infected”

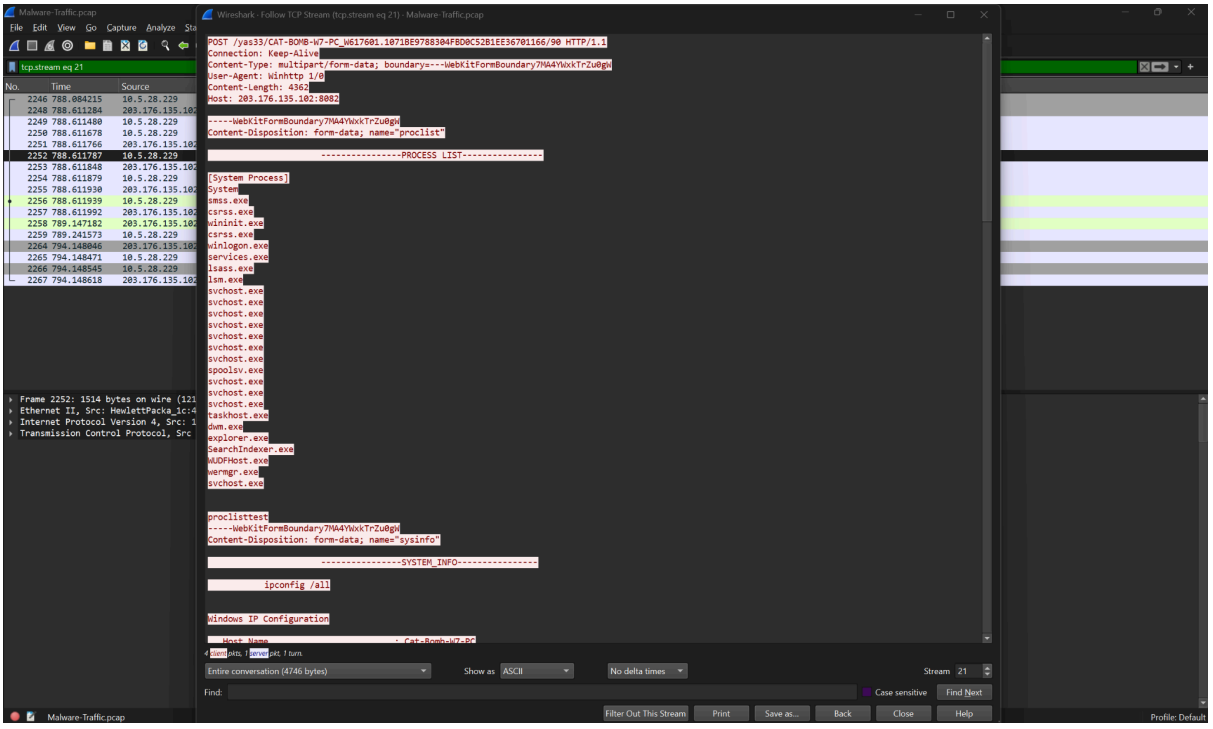
Step2: Launch Wireshark. From the main menu select “File,” select “Open,” and then open the file namedMalwareTraffic-1.pcap.



When Trick bot successfully infects a Windows host, it sends an HTTP POST request with the system data, usually over TCP port 8082. The URL ends with /90, so use the following Wireshark filter to find that URL and follow the TCP stream:
http.request.uri contains "/90"
This should return two URLs in your Wireshark column display, one for the infected Windows client (CAT-BOMB-W7-PC), and one for the domain controller (CATBOMBER-DC).



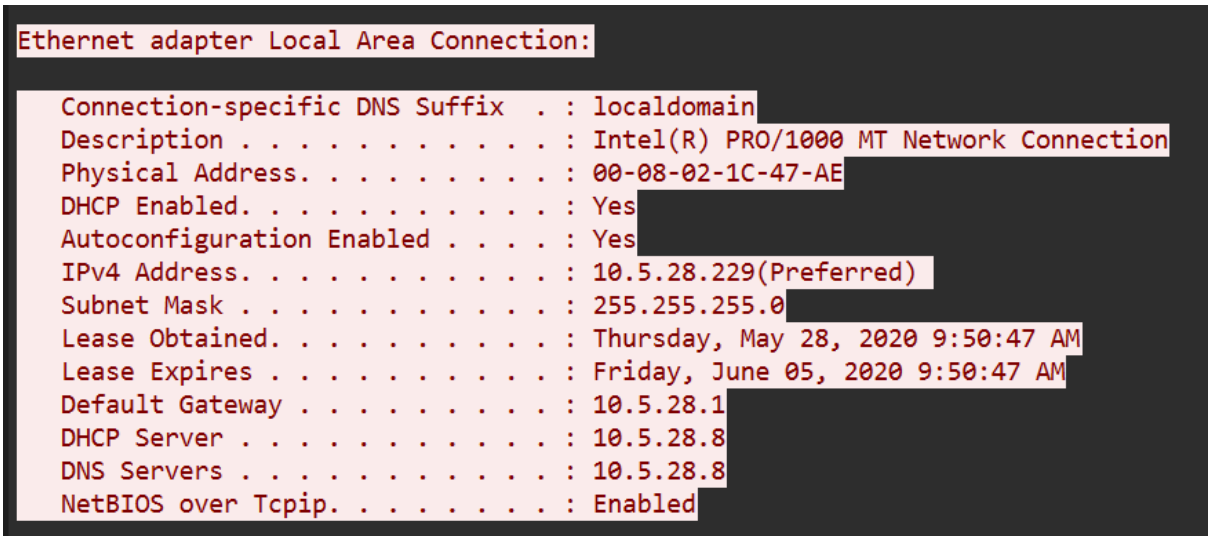
9) Now we right click the first packet, numbered 2256, and select follow, then TCP stream.



Explanation:
Trickbot malware transmits system details using HTTP POST requests. Examining the TCP stream allows us to determine the infected machine, its hostname, and the related user accounts.

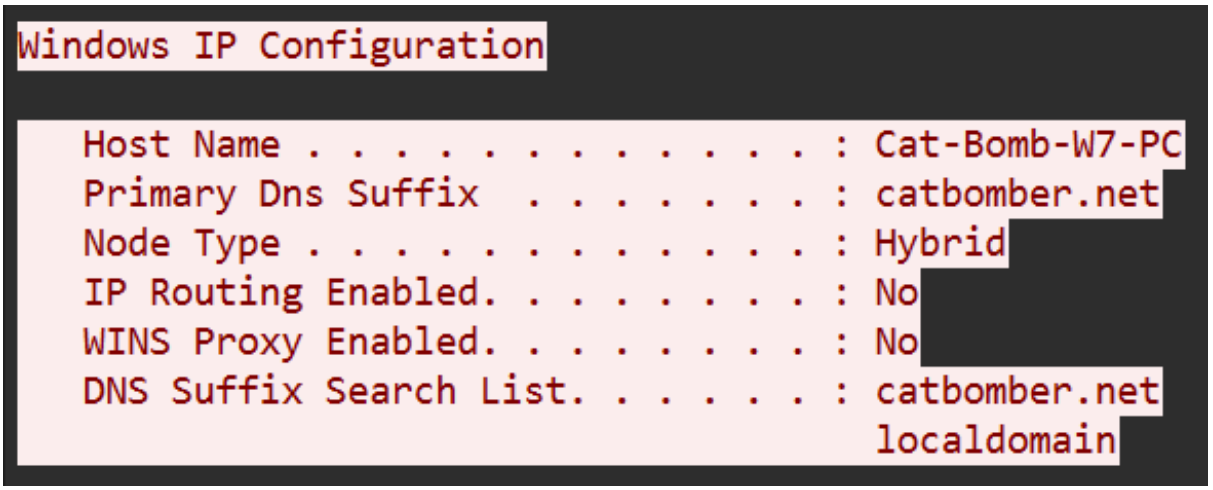
Question 1. Answer the following and provide the screenshots:

Infected Windows client IP address:



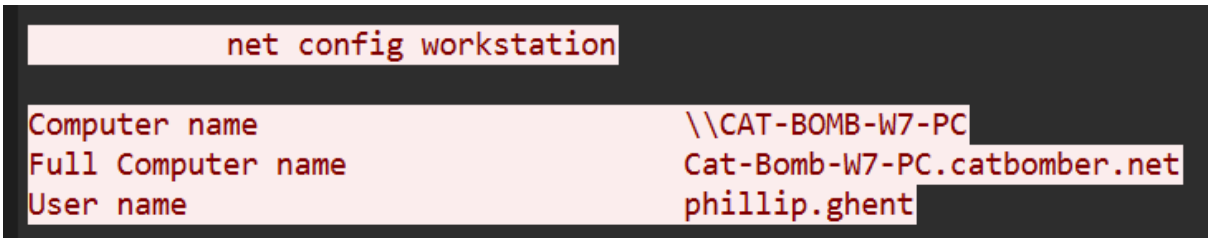
Answer: The Infected Windows client IP address is 10.5.28.229

Infected Windows client host name:



Answer: The Infected Windows client host name is CAT-BOMB-W7-PC

Infected Windows client user account name:



Answer: The Infected Windows client user account name is phillip.ghent

Other Windows client host name:
Answer: The Other Windows client host name is CAT-BOMB-W10-PC

```
Domain: Catbomber-DC.catbomber.net

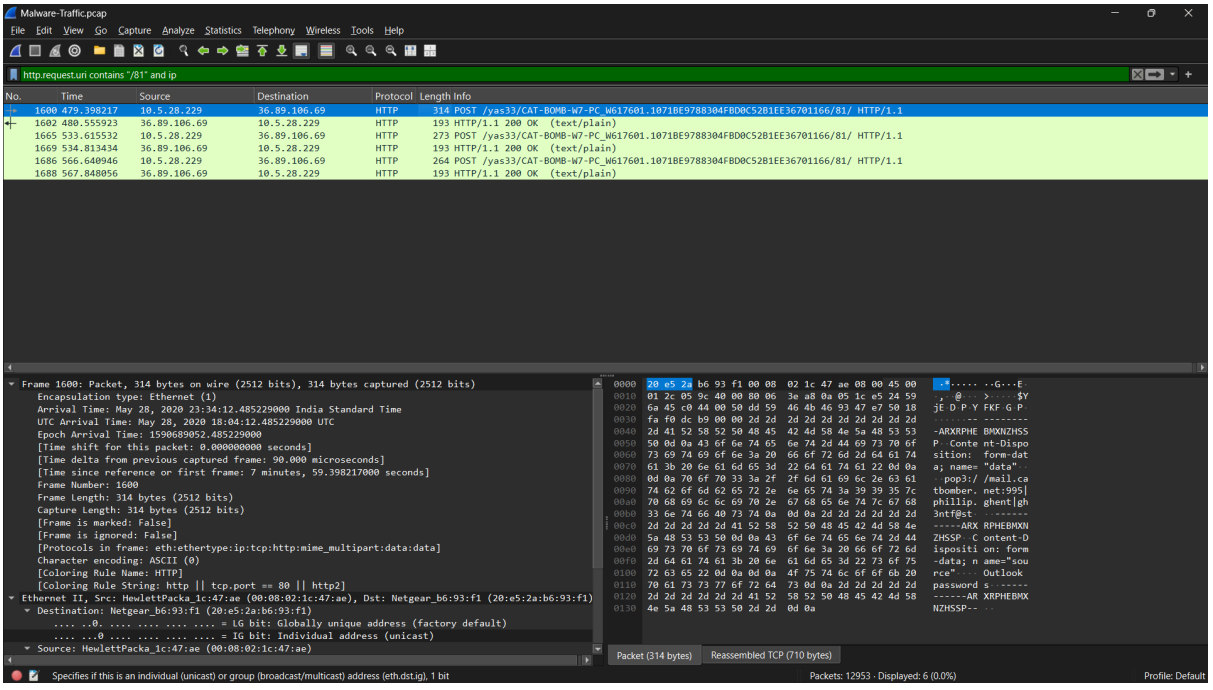
Name: Catbomber-DC.catbomber.net
Name: CAT-BOMB-W10-PC.catbomber.net
Name: CAT-BOMB-W7-PC.catbomber.net
```

Other Windows client user account name:
Answer: The Other Windows client user account name is timothy.sizemore

```
Username: Administrator Username: Guest Username: krbtgt Username: timothy.sizemore Username: phillip.ghent -
----
```

Step3: Click the “Close” button to close the “Follow TCP stream window”. Click the X button, at the end of the textbox to clear the filter, which was automatically applied.

Step4: HTTP POST requests that end in "/81" is where we find password data exfiltrated from an infected Windows host. Use the following Wireshark filters to find email passwords: http.request.uri contains "/81" and ip



Step5: Right-click on the packet numbered 1600 and from the pull-down menu select “Follow”,and then “TCP Stream”



```
POST /yas33/CAT-BOMB-W7-PC_W617601.1071BE9788304FBD0C52B1EE36701166/81/ HTTP/1.1
Accept: */*
Content-Type: multipart/form-data; boundary=---------ARXRPHEBMXNZHSSP
Connection: Close
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; Win64; x64; Trident/7.0; .NET CLR 2.0.50727; SLCC2; .NET CLR 3.5.30729; .NE
T CLR 3.0.30729; Media Center PC 6.0; .NET4.0C; .NET4.0E)
Host: 36.89.106.69
Content-Length: 260
Cache-Control: no-cache

-----------ARXRPHEBMXNZHSSP
Content-Disposition: form-data; name="data"

pop3://mail.catbomber.net:995|phillip.ghent|gh3ntf@st

-----------ARXRPHEBMXNZHSSP
Content-Disposition: form-data; name="source"

Outlook passwords
-----------ARXRPHEBMXNZHSSP--

HTTP/1.1 200 OK
connection: close
server: Cowboy
date: Thu, 28 May 2020 18:04:12 GMT
content-length: 3
Content-Type: text/plain

/1/
```

Question 2. Answer the following Questions:

What is user Philip’s password?



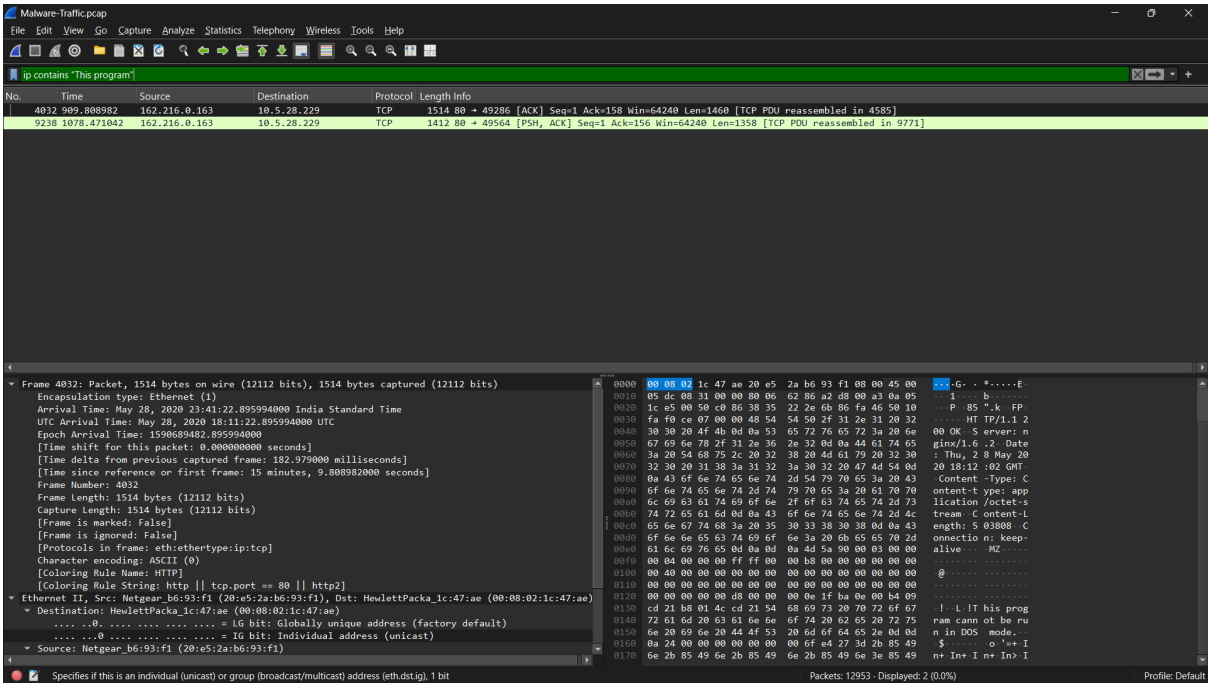
```
pop3://mail.catbomber.net:995|phillip.ghent|gh3ntf@st
```

Answer: The infected User Philip Ghents email password is gh3ntf@st

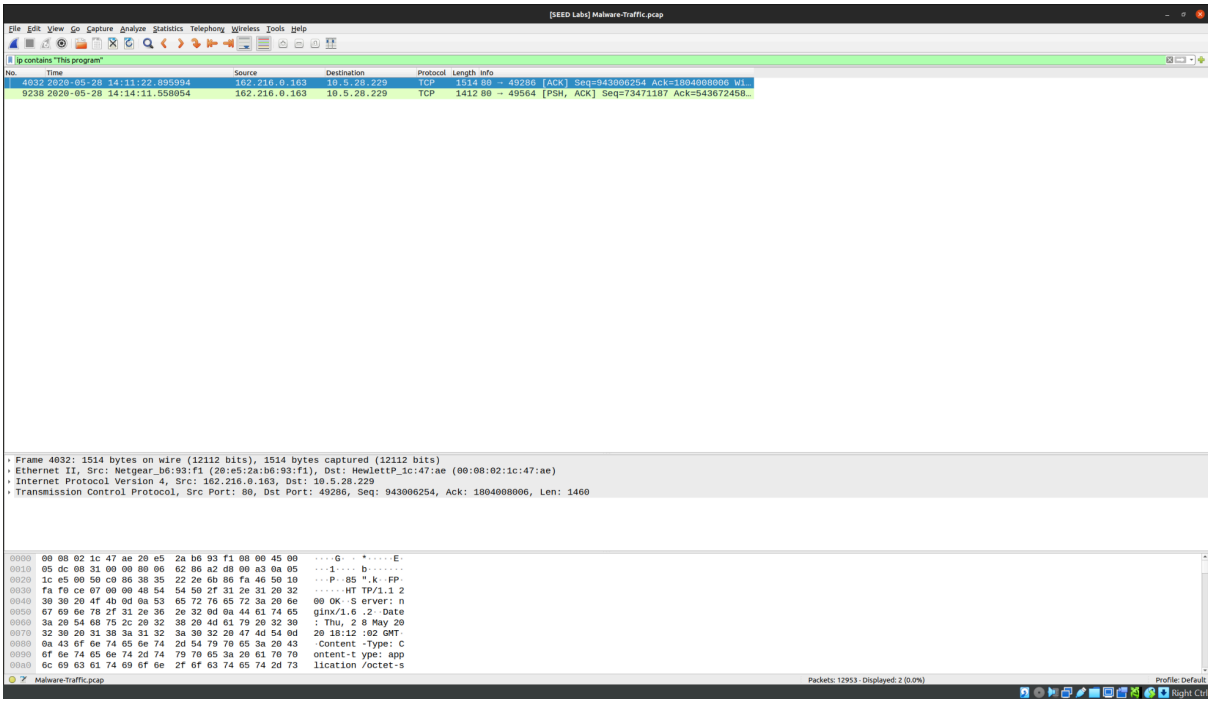
Step6: Click the “Close” button to close the “Follow TCP stream window”. Click the X button, at the end of the textbox to clear the filter, which was automatically applied. We can quickly filter on traffic to see if any Windows executable (EXE) files pass in the clear (not as encoded or encrypted data) using the following filter:

ip contains "This program"

Note: This doesn't work every single time, but it works for most EXE files. It should return two frames in your column display. Follow each of these TCP streams.



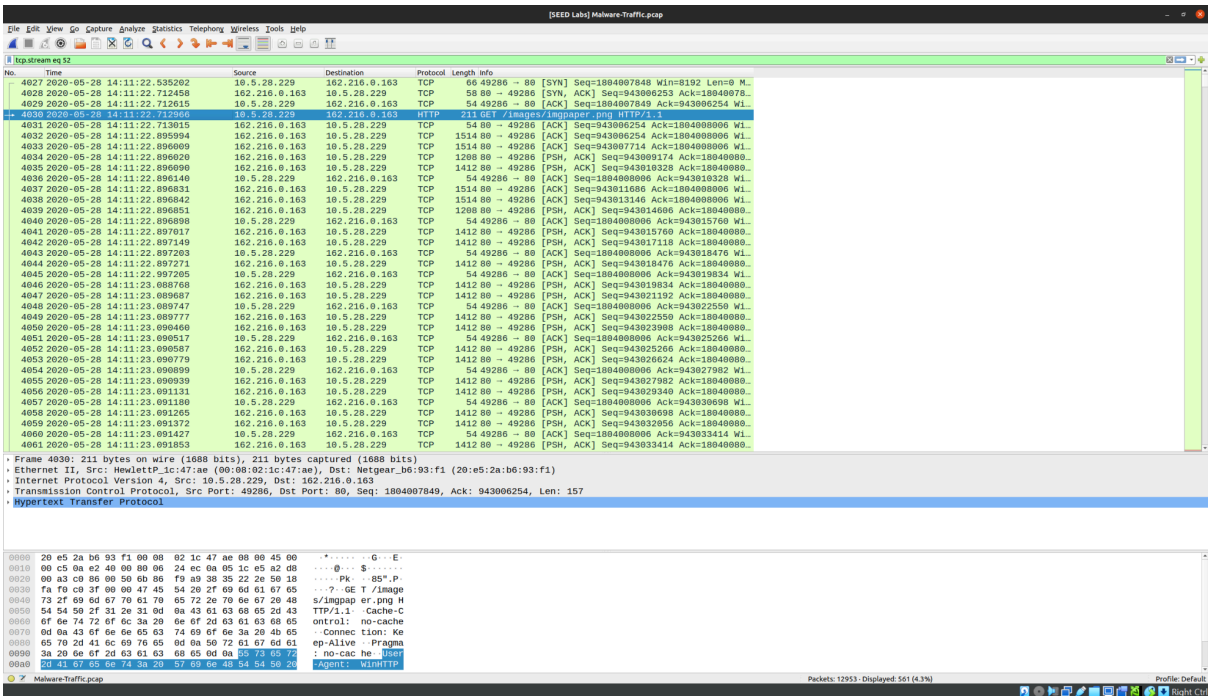
Note: Windows defender removes the files automatically due to security. Hence the last part of the lab was performed on ubuntu vm.



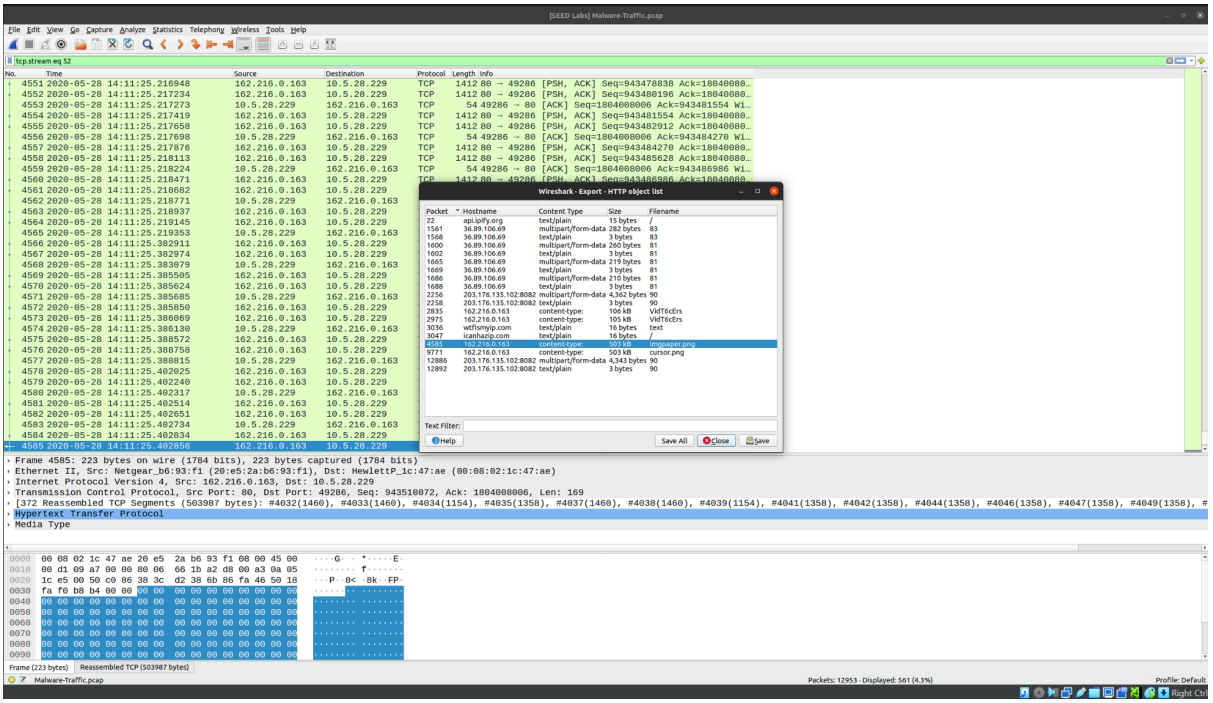
Step7: Follow TCP stream of packets:
Packet 4032:

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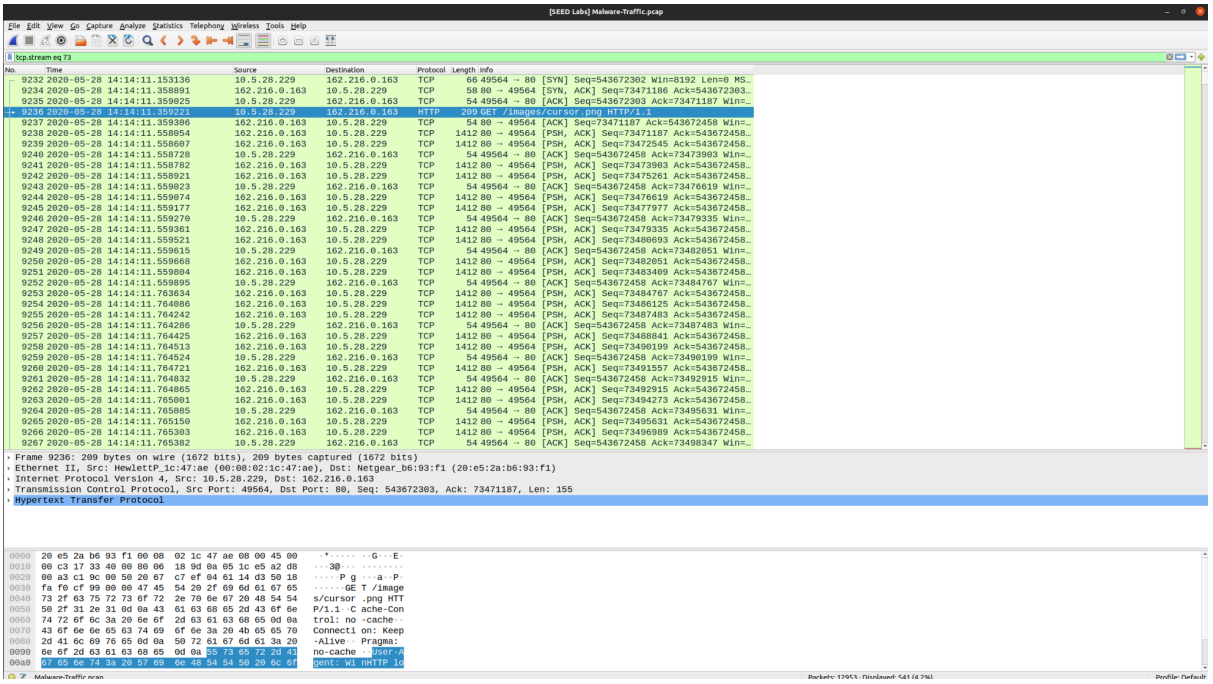


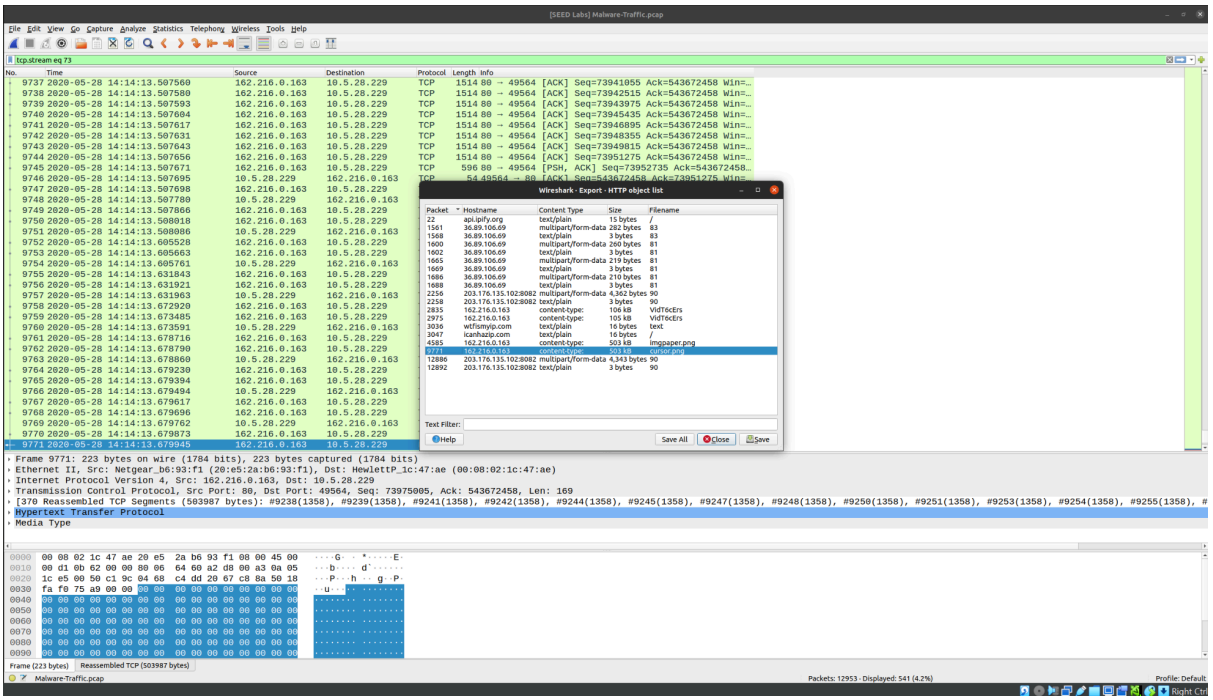
Jan - May 2026 Assignment SUBMISSION_UE23CS342BA6



Packet 9238:







Step8: Export the executables:
Goto File/Export Objects/HTTP and save the files



Step9: Calculate the sha256 hash

10)Command: sha256sum filename.exe

Executable 1 (cursor.png) SHA256:
4e76d73f3b303e481036ada80c2eeba8db2f306cbc9323748560843c80b2fed1

Executable 2 (imgpaper.png) SHA256:
934c84524389ecfb3b1dfcb28f9697a2b52ea0ebcaa510469f0d2d9086bcc79a